



Fr. C. Rodrigues Institute of Technology, Vashi

(An Autonomous Institute & Permanently Affiliated to University of Mumbai)

Department of Computer Engineering

Sem/Div : VII/A

Department of Computer Engineering

Time: 1 Hour

Subject : Natural Language Processing Assignment Test -I (SH-2025)

Total Marks: 20

SET B

Q.	All questions are compulsory.(Each question carries 05 marks)	CO	PO	PSO	BTL
a)	List and define the various stages of Natural Language Processing (NLP).	1	1,2, 12	1,2	1
b)	Consider the following corpus • <s> Book/V a/DET car/N </s> • <s> Park/V the/DET car/N </s> • <s> The/DET book/N is/V in/Prep the/DET car/N </s> • <s> The/DET car/N is/V in/Prep a/DET park/N </s> N: Noun, DET: Determiner ,V: Verb, Prep:Preposition Compute the emission and transition probabilities for a bigram HMM.	3	1,2,5,15	1,2	3
c)	Compute the perplexity of the sentence “She likes mangoes” using a bigram language model constructed from the given text corpus: She likes apples.She likes bananas.He likes mangoes.They eat fruits.She likes grapes.	2	1,2,5,15	1,2	3
d)	Consider the grammar $S \rightarrow VP PP$ $VP \rightarrow DT VB$ $PP \rightarrow Prep NP$ $NP \rightarrow DT NN$ $NN \rightarrow JJ NN$ $DT \rightarrow The A$ $Prep \rightarrow of in$ $VB \rightarrow Whispering Killing Screaming Sacrifice Haunting$ $JJ \rightarrow Mysterious Ominous Dark Bleeding Possessed$ $NN \rightarrow Ghoul Castle Witch Vampire Forest$ Compute the CKY table to check the string “The Sacrifice of a dark vampire”, belongs to the given grammar.	3	1,2,5,15	1,2	3